

Business Understanding

1. Introduction

The purpose of this project is to analyze Amazon sales data and extract meaningful business insights that help improve business performance. The project follows an end-to-end Data Analytics workflow including business understanding, SQL analysis, Python data cleaning, exploratory data analysis (EDA), data visualization, Power BI dashboard development, AI-assisted business insights, and business recommendations.

The main objective is to understand customer purchasing behavior, sales trends, product performance, payment preferences, and regional performance to support data-driven business decisions.

2. Dataset Overview

The dataset contains transactional sales records of an Amazon-like e-commerce platform. Each record represents a single customer order and includes customer information, product details, pricing, quantity, discounts, taxes, shipping costs, payment methods, seller details, and order status. The dataset is suitable for performing sales analytics, customer behavior analysis, revenue analysis, and business performance evaluation.

3. Business Problem

An e-commerce company generates thousands of orders per day. Without proper analysis, management cannot identify sales trends,

customer preferences, product performance, or operational issues.

The company needs an analytics solution that can answer important business questions such as:

- Which products generate the highest revenue?
- Which brands perform best?
- Which payment methods are preferred by customers?
- Which states generate the highest sales?
- How effective are discounts?
- Which products should be promoted?
- Which areas need business improvement?

The objective is to transform raw sales data into meaningful insights that support better business decisions.

4. Business Objectives

The objectives of this project are:

- Analyze overall sales performance.
- Identify top-performing products and categories.
- Analyze customer purchasing behavior.
- Identify the most preferred payment methods.
- Measure the impact of discounts on sales.
- Evaluate regional sales performance.

- Identify best-performing brands.
- Monitor order status and cancellations.
- Build an interactive Power BI dashboard.
- Generate AI-assisted business insights.
- Provide actionable business recommendations.

5. Stakeholders

The stakeholders involved in this project include:-

Stakeholder	Responsibility
Executive Management	Monitor overall business performance and make strategic decisions
Sales Team	Track sales performance and identify high-performing products
Marketing Team	Plan promotional campaigns based on customer behavior
Finance Team	Monitor revenue, discounts, taxes, and profitability
Inventory Management Team	Maintain stock levels based on demand
Logistics & Delivery Team	Monitor shipping costs and delivery performance
Sellers/Vendors	Evaluate product sales and business performance
Customers	Purchase products and influence sales trends
Business Analysts	Analyze data and generate business insights

6. Column Description

Column	Business Meaning
OrderID	Unique identifier for each order
OrderDate	Date when the customer placed the order
CustomerID	Unique customer identifier
CustomerName	Name of the customer
ProductID	Unique product identifier
ProductName	Name of the product
Category	Product category
Brand	Product brand
Quantity	Number of units purchased
UnitPrice	Price of one unit
Discount	Discount applied
Tax	Tax charged
ShippingCost	Delivery cost
TotalAmount	Final amount paid by customer
PaymentMethod	Customer payment method
OrderStatus	Order status (Delivered, Cancelled, etc.)
City	Customer city
State	Customer state
Country	Customer country
SellerID	Seller identifier

7. Dataset Description

Each row in the dataset represents one customer order transaction.

The dataset contains information related to:

- Customer Details
- Product Details
- Pricing
- Quantity Purchased
- Discounts
- Taxes
- Shipping Charges
- Payment Method
- Delivery Status
- Seller Information
- Geographic Location

8. Business Questions

The following business questions will be answered during the analysis:

1. Sales Analysis

- What is the total revenue generated?
- Which product category generates the highest revenue?
- Which products are sold the most?

- Which brands perform the best?
- What are the monthly sales trends?
- Which products generate the highest sales value?

2. Customer Analysis

- How many unique customers are there?
- Which customers purchase most frequently?
- What is the average order value?
- Which payment methods are most preferred?

3. Regional Analysis

- Which countries generate the highest revenue?
- Which states generate the highest sales?
- Which cities contribute the most revenue?

4. Operational Analysis

- How many orders are delivered?
- How many orders are cancelled?
- What is the cancellation rate?
- What is the average shipping cost?
- How much discount is provided overall?

5. Financial Analysis

- Which categories receive the highest discounts?
- Which brands generate maximum revenue?

- What is the average tax collected?
- What is the average revenue per order?

9. Key Performance Indicators (KPIs)

- Revenue KPIs :-
 1. Total Revenue
 2. Average Order Value
 3. Revenue by Category
 4. Revenue by Brand
 5. Revenue by State
 6. Revenue by City
- Sales KPIs :-
 7. Total Orders
 8. Total Quantity Sold
 9. Best Selling Product
 10. Best Selling Category
- Customer KPIs:-
 12. Total Customers
 13. Repeat Customers
 14. Average Products per Order
 15. Preferred Payment Method

- Operational KPIs :-

16.Delivered Orders

17.Cancelled Orders

18.Cancellation Rate

19.Average Shipping Cost

20.Average Discount

10. Expected Business Outcomes

After completing this project, the business will be able to:

- Identify high-performing products and brands.
- Understand customer purchasing patterns.
- Monitor regional sales performance.
- Improve inventory planning.
- Optimize pricing and discount strategies.
- Improve delivery performance.
- Support data-driven decision-making through interactive dashboards and business reports.